

- Random Sampling
- Stratified Random Sampling
- Systematic Sampling
- Cluster Sampling

Normal Distribution:

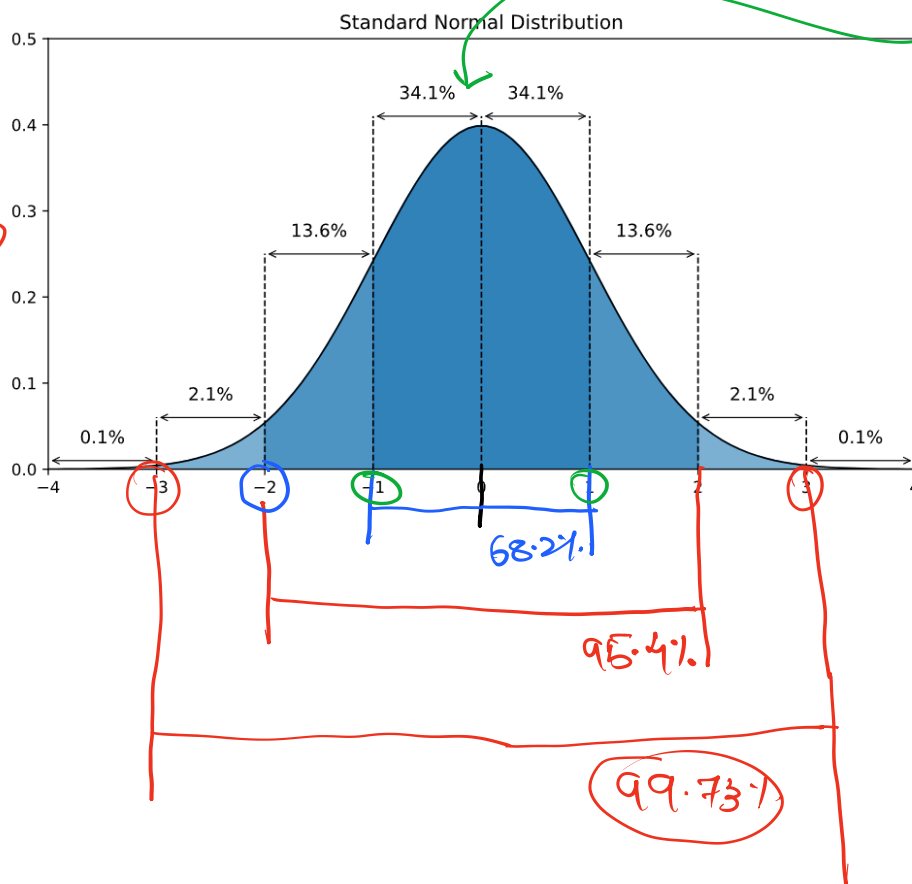
z^* → critical value

sample data
↓
kdeplot

z-scores

Mean - 1SD

$z = -2$



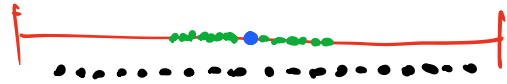
-1 to $+1$
68.2%

SD = 5

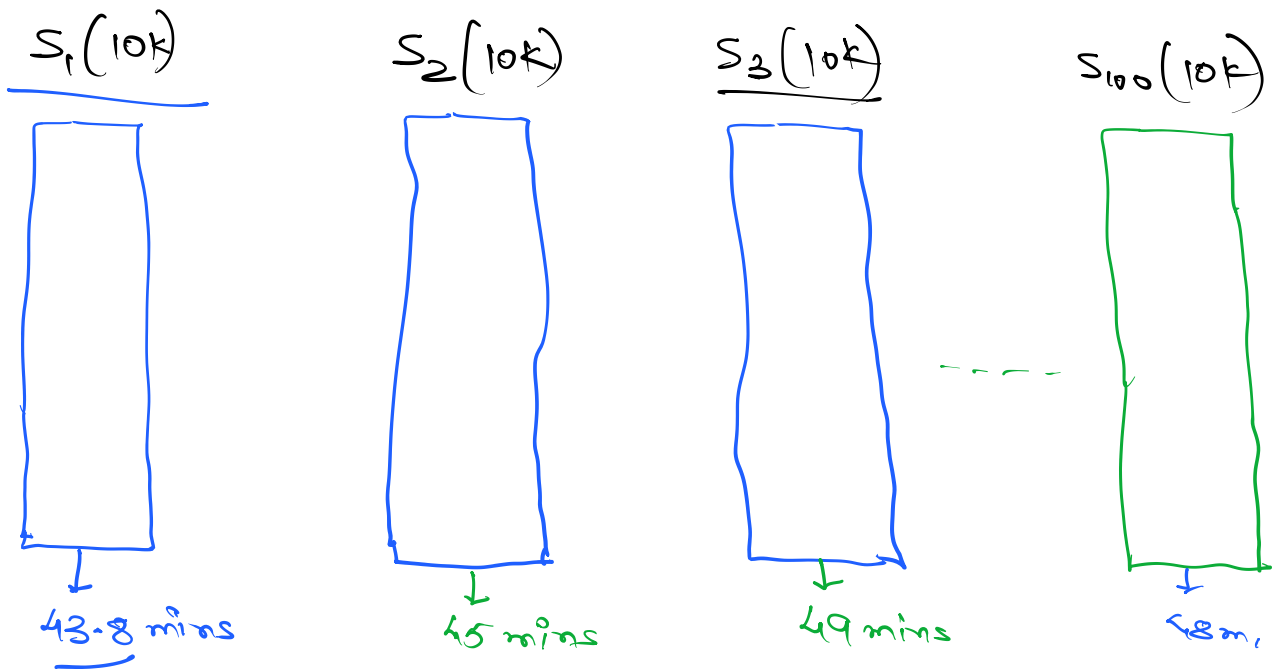
1 km
↓
1000 m

$$\begin{aligned} \text{Margin of Error} &= z^* \times \text{Standard Error} \\ &= z^* \times \frac{s}{\sqrt{n}} \end{aligned}$$

Standard Error:



SD: How spread out the values are from the mean of the data



Std Error: Std dev of Sample Means

$[s_{19,1}, s_{19,2}, \dots, s_{19,100}]$

Pop SD

$$\text{Std error: } \frac{\text{Sample SD}}{\sqrt{n}}$$

1 sample
↓
SD
SS

